Droop Adjust Signal Malfunctions

Symptoms

- Engine does **not** respond to droop adjust request.

How To Use This Tree

This symptom tree can be used to troubleshoot droop adjust symptoms. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

None.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the customer interface box (C.I.B.) wiring.	
	STEP 1A. Check the droop adjust potentiometer SUPPLY wire.	Less than 10 ohms?
	STEP 1B. Check the droop adjust potentiometer RETURN wire.	Less than 10 ohms?
	STEP 1C. Check the droop adjust potentiometer SIGNAL wire.	Less than 10 ohms?
STEP 2.	Check the engine harness to the C.I.B. cable.	
	STEP 2A. Check the droop adjust potentiometer SUPPLY and SIGNAL wires.	Less than 10 ohms?
	STEP 2B. Check the droop adjust potentiometer RETURN and SIGNAL wires.	Less than 10 ohms?

STEP 1. Check C.I.B. wiring.

STEP 1A. Check the droop adjust potentiometer SUPPLY wire.

Conditions :

- Open the C.I.B.
- Disconnect the C.I.B. to the engine harness cable connector C1 from the C.I.B.

Action	Specification/Repair	Next Step
Check the droop adjust potentiometer SUPPLY wire. • Place one test lead on the droop adjust potentiometer 5 volt SUPPLY pin in connector 1. • Place the other test lead on the droop adjust potentiometer 5 volt SUPPLY terminal on the X1 connector.	Less than 10 ohms? **YES**	**1B**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	**Repair complete**

STEP 1B. Check the droop adjust potentiometer RETURN wire.

Conditions :

- Open the C.I.B.
- Disconnect the C.I.B. to the engine harness cable connector C1 from the C.I.B.

Action	Specification/Repair	Next Step

Check the droop adjust potentiometer RETURN wire. • Place one test lead on the droop adjust potentiometer RETURN pin in connector C1. • Place the other test lead on droop adjust potentiometer RETURN terminal on the X1 connector.	Less than 10 ohms? **YES**	**1C**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	**Repair complete**

STEP 1C. Check the droop adjust potentiometer SIGNAL wire.

Conditions :

- Open the C.I.B.
- Disconnect the C.I.B. to the engine harness cable connector C1 from the C.I.B.

Action	Specification/Repair	Next Step
Check the droop adjust potentiometer SIGNAL wire. • Place one test lead on the droop adjust potentiometer SIGNAL pin in connector C1. • Place the other test lead on droop adjust potentiometer signal terminal on the X1 connector.	Less than 10 ohms? **YES**	**2A**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	**Repair complete**

STEP 2. Check the engine harness to the C.I.B. cable.

STEP 2A. Check the droop adjust potentiometer SUPPLY and SIGNAL wires.

Conditions :

- Disconnect the C.I.B. to the engine harness cable connector C1 from the C.I.B.
- Disconnect the C.I.B. to the engine harness cable from the engines.

Action	Specification/Repair	Next Step
Check the droop adjust potentiometer SUPPLY and SIGNAL wires. • Place a jumper between the droop adjust potentiometer 5 volt SUPPLY pin and the droop adjust potentiometer SIGNAL pin in the engine-side connection. • Place one test lead in the droop adjust potentiometer 5 volt SUPPLY pin of the C1 connector. • Place the other test lead in the droop adjust potentiometer SIGNAL pin of the C1 connector.	Less than 10 ohms? **YES**	**2B**
	Less than 10 ohms? **NO** **Repair :** Replace the cable.	**Repair complete**

STEP 2B. Check the droop adjust potentiometer RETURN and SIGNAL wires.

Conditions :

- Disconnect the C.I.B. to the engine harness cable connector C1 from the C.I.B.
- Disconnect the C.I.B. to the engine harness cable from the engine.

Action	Specification/Repair	Next Step

Check the droop adjust potentiometer RETURN and SIGNAL wires. • Place a jumper between the droop adjust potentiometer SIGNAL pin and the droop adjust potentiometer RETURN pin in the engine-side connection. • Place one test lead in the droop adjust potentiometer RETURN pin of the C1 connector. • Place the other test lead in the droop adjust potentiometer SIGNAL pin of the C1 connector.	Less than 10 ohms? **YES** **Repair :** Use the following manuals for potentiometer repair instructions. Marine Auxiliary QSB7-DM CM850 Fault Code Troubleshooting Manual, Bulletin 4325972, Section TF; ISM and QSM11 Electronic Control System Troubleshooting and Repair Manual, Bulletin 3666266, Section TF; X15 CM2350 X125M Fault Code Troubleshooting Manual, Bulletin 5504346, Section TF; or the original equipment manufacturer (OEM) service manual.	**Repair complete**
	Less than 10 ohms? **NO** **Repair :** Replace the cable.	**Repair complete**